

Nursing Knowledge — facts for clinical review

NRG Platform · 133 facts · generated 20 September 2026

What this is, and what is being asked

These are the prompts shown in the floating **Nursing Knowledge** card on the NRG platform. Students see the *prompt* first and tap to reveal the *answer*.

Please confirm each one is clinically correct and appropriate for RENR candidates. Tick the OK box, or write a correction in the right-hand column. Anything left unticked will be treated as not yet approved.

Two notes on the content. Reference ranges are given in **SI units** (mmol/L, g/L) to match Caribbean and UK laboratory reporting, and they vary slightly between laboratories — the intent is to teach the shape of a value, not a local reference range. Where a number is a clinical threshold rather than a range, please check it against current local guidance.

These facts were drafted to be standard, textbook-level and high-yield. They have **not** yet had a nursing sign-off, which is what this document is for.

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Vital Signs

10 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
1	☐	Normal adult vitals you must know	T 36.5–37.5°C · P 60–100 bpm · R 12–20/min · BP <120/80 · SpO ₂ ≥95%	
2	☐	Normal newborn heart and respiratory rate	HR 110–160 bpm · RR 30–60/min. Both are far faster than an adult's — count for a full minute.	
3	☐	What counts as a hypertensive crisis?	BP ≥180/120 mmHg. Treat as an emergency and look for end-organ signs: headache, visual change, chest pain.	
4	☐	How is orthostatic hypotension defined?	A fall of ≥20 mmHg systolic or ≥10 mmHg diastolic within 3 minutes of standing. Check lying, sitting, standing.	
5	☐	Mean arterial pressure — the number that matters for perfusion	MAP ≈ (systolic + 2 × diastolic) ÷ 3. Organs need roughly ≥65 mmHg to perfuse.	
6	☐	Cushing's triad signals what?	Rising intracranial pressure: hypertension with a widening pulse pressure, bradycardia, and irregular respirations. A late and ominous sign.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
7	☐	Minimum acceptable adult urine output	0.5 mL/kg/hour, or roughly 30 mL/hour. Less than that is oliguria — report it.	
8	☐	Why take an apical pulse for a full minute?	Because an irregular rhythm cannot be extrapolated from 15 seconds. Always a full minute before digoxin and in any arrhythmia.	
9	☐	Pulse pressure — and when it narrows	Systolic minus diastolic, normally 30–40 mmHg. A narrowing pulse pressure can be an early shock sign.	
10	☐	Fever in an older adult	May be absent. An older person with infection often presents with confusion, falls or reduced intake rather than a temperature.	

Lab Values

17 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
11	☐	Serum potassium — the classic RENR value	3.5–5.0 mmol/L. Below 3.5 → hypokalaemia: flattened T waves, weakness, arrhythmia.	
12	☐	Serum sodium	135–145 mmol/L. Sodium problems present neurologically: confusion, headache, seizures.	
13	☐	Total serum calcium	2.1–2.6 mmol/L. Low calcium → tingling, tetany, Trousseau's and Chvostek's signs.	
14	☐	Serum magnesium	0.7–1.0 mmol/L. Low magnesium makes hypokalaemia refractory — replace magnesium or the potassium won't correct.	
15	☐	Fasting blood glucose	3.9–5.5 mmol/L normally. Fasting ≥ 7.0 mmol/L on two occasions supports a diabetes diagnosis.	
16	☐	HbA1c — what it tells you	Average glycaemia over about 3 months. $\geq 6.5\%$ is diagnostic of diabetes; most adults aim below 7%.	
17	☐	Haemoglobin reference range	Roughly 130–170 g/L in men and 120–155 g/L in women. Below that is anaemia — look for the cause.	
18	☐	Platelet count	150–400 $\times 10^9$ /L. Below 50 raises real bleeding risk; below 20 is spontaneous bleeding territory.	
19	☐	White cell count	4–11 $\times 10^9$ /L. A left shift (more neutrophils) suggests bacterial infection; a very low count means neutropenic precautions.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
20	☐	Therapeutic INR on warfarin	Usually 2.0–3.0. Above range means bleeding risk — the antidote is vitamin K.	
21	☐	Therapeutic digoxin level	0.5–2.0 ng/mL. Above 2.0 is toxic: nausea, visual halos or yellow vision, bradycardia, confusion.	
22	☐	Therapeutic lithium level	0.6–1.2 mmol/L — a narrow window. Above 1.5 is toxic: tremor, vomiting, ataxia, confusion.	
23	☐	Troponin timing after myocardial infarction	Begins rising about 3–4 hours after injury and peaks within about 24 hours. A single early normal value does not exclude an MI.	
24	☐	What does a raised BNP suggest?	Ventricular stretch — most often heart failure. It helps separate cardiac from respiratory breathlessness.	
25	☐	Potassium above 6.5 mmol/L	A medical emergency. Expect peaked T waves and a risk of fatal arrhythmia. Get an ECG and stop all potassium.	
26	☐	Serum creatinine — what it reflects	Kidney function. A rising creatinine with falling urine output is acute kidney injury; it also changes drug dosing.	
27	☐	Reading a trend versus a single value	A value inside range that has moved sharply can matter more than one just outside it. Always compare with the last result.	

Medication Safety

20 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
28	☐	The 5 Rights of medication administration	Right patient · right drug · right dose · right route · right time. Two identifiers, every time.	
29	☐	When do you hold digoxin?	If the adult apical pulse is below 60 bpm. Withhold, recheck, and report before giving.	
30	☐	Why does hypokalaemia matter on digoxin?	Low potassium potentiates digoxin toxicity. Watch the potassium of anyone on digoxin and a diuretic.	
31	☐	Which tablets must never be crushed?	Enteric-coated and extended- or sustained-release forms. Crushing them delivers the whole dose at once.	
32	☐	Mixing insulins in one syringe	Draw clear before cloudy — regular before NPH — so the short-acting vial is never contaminated.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
33	☐	Rapid-acting insulin and mealtimes	Onset is about 15 minutes. Food must be in front of the patient before you give it.	
34	☐	Antidotes worth memorising (1)	Warfarin → vitamin K · Heparin → protamine sulfate · Opioids → naloxone.	
35	☐	Antidotes worth memorising (2)	Paracetamol/acetaminophen → N-acetylcysteine · Benzodiazepines → flumazenil · Magnesium → calcium gluconate.	
36	☐	Giving IV potassium	Never as a bolus or push. Always diluted, always on a pump, and watch the site — it is intensely irritant.	
37	☐	Classic ACE inhibitor adverse effects	A dry persistent cough, hyperkalaemia, and — rarely but dangerously — angioedema. Stop and escalate for airway swelling.	
38	☐	Why are beta blockers risky in diabetes?	They mask the adrenergic warning signs of hypoglycaemia — tremor and tachycardia. Sweating usually remains.	
39	☐	Never stop a beta blocker abruptly	Abrupt withdrawal can cause rebound tachycardia, hypertension and ischaemia. Taper under supervision.	
40	☐	Furosemide — what to monitor	Potassium, sodium and hydration. Push slowly: rapid IV administration risks ototoxicity.	
41	☐	Metformin and contrast imaging	Usually withheld around iodinated contrast because of the lactic acidosis risk if kidney function drops.	
42	☐	Aminoglycosides such as gentamicin	Nephrotoxic and ototoxic. Monitor trough levels, creatinine, and ask about hearing or balance change.	
43	☐	MAOIs and tyramine	Aged cheese, cured meat, red wine and soy can trigger a hypertensive crisis. Strict dietary teaching is essential.	
44	☐	Which medicines are 'high alert'?	Insulin, heparin and anticoagulants, opioids, and concentrated potassium. Most require an independent double-check.	
45	☐	A patient questions their medication	Stop. Do not give it. Re-verify against the order — patients catch real errors.	
46	☐	Verbal orders	Write it down and read it back to the prescriber. Never act on an unrepeatable verbal order.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
47	☐	Nitroglycerin before you give it	Check the blood pressure. Typically withheld if systolic is below 90 mmHg. Headache is expected, not an allergy.	

Prioritisation

10 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
48	☐	The first framework for any 'what do you do first?'	ABC — airway, then breathing, then circulation. An airway problem always outranks everything else.	
49	☐	Where does Maslow fit in prioritisation?	Physiological needs before psychosocial ones. Pain and oxygen come before reassurance.	
50	☐	Actual versus potential problems	An actual problem outranks a potential one — unless the potential one is airway, breathing or circulation.	
51	☐	Which patient do you see first?	The unstable one, the acute change over the chronic state, and the unexpected finding over the expected one.	
52	☐	When the options are assess or intervene	Assess first, unless the situation is immediately life-threatening. You cannot treat what you have not established.	
53	☐	A patient is in immediate danger — do you leave to get help?	No. Stay and call for help from the room. Never leave an unsafe patient unattended.	
54	☐	What can never be delegated by a registered nurse?	Assessment, teaching, evaluation, clinical judgement, and the care of an unstable patient.	
55	☐	Delegating to unlicensed assistive personnel	Suits stable, predictable, routine tasks with a clear expected outcome — and you remain accountable for the outcome.	
56	☐	A trend versus a snapshot	Three readings drifting the wrong way beat one abnormal number. Always look back before you act.	
57	☐	The single most reliable early warning sign	A change in level of consciousness or new confusion. It precedes most deterioration.	

Infection Control

11 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
58	☐	The single most effective infection control measure	Hand hygiene, before and after every patient contact. Nothing else comes close.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
59	☐	Who gets standard precautions?	Every patient, every time, regardless of diagnosis. They assume all blood and body fluid is infectious.	
60	☐	Airborne precautions — which conditions and what PPE?	Tuberculosis, measles, varicella. N95 respirator and a negative-pressure room with the door closed.	
61	☐	Droplet precautions	Influenza, pertussis, meningococcal disease. A surgical mask within about a metre, and a private room if possible.	
62	☐	Contact precautions	MRSA, C. difficile, scabies. Gown and gloves on entry, dedicated equipment.	
63	☐	Why soap and water for C. difficile?	Alcohol gel does not kill spores. Wash with soap and water, and clean the room with a sporicidal agent.	
64	☐	Order for putting PPE on	Gown, mask or respirator, eye protection, then gloves.	
65	☐	Order for taking PPE off	Gloves, eye protection, gown, then mask — leaving the most contaminated item first and the airway protection last.	
66	☐	Sterile field rules that catch people out	Anything below waist level is contaminated, a 2.5 cm border is unsterile, and moisture wicks contamination through.	
67	☐	Caring for a neutropenic patient	Protective precautions: no unwashed raw food, no fresh flowers or standing water, and no visitors with any infection.	
68	☐	Needle safety	Never recap a used needle. Dispose at the point of use, in a sharps container you can see.	

Fluids & Electrolytes

8 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
69	☐	Name the isotonic fluids	0.9% sodium chloride and lactated Ringer's. They stay in the extracellular space and expand volume.	
70	☐	What is the risk with hypotonic fluid?	0.45% saline pulls water into cells — dangerous where cerebral oedema is a concern.	
71	☐	Signs of hypovolaemia	Dry mucous membranes, poor skin turgor, tachycardia, falling BP, concentrated urine, thirst.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
72	☐	Signs of fluid overload	Crackles, raised jugular venous pressure, bounding pulse, peripheral oedema, sudden weight gain.	
73	☐	The quickest fluid-balance measure at the bedside	Daily weight, same scale, same time, same clothing. About 1 kg gained equals about 1 litre retained.	
74	☐	Emergency treatment of hyperkalaemia	Calcium to stabilise the myocardium, insulin with dextrose to shift potassium into cells, then something to remove it.	
75	☐	Correcting sodium — why slowly?	Rapid correction of chronic hyponatraemia can cause osmotic demyelination. Slow and monitored.	
76	☐	Two bedside signs of hypocalcaemia	Trousseau's sign — carpal spasm with a BP cuff — and Chvostek's sign, facial twitch on tapping the facial nerve.	

Respiratory

6 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
77	☐	Oxygen target in COPD	Usually 88–92%. Over-oxygenating a CO ₂ retainer can suppress the drive to breathe.	
78	☐	The most ominous sign in acute asthma	A silent chest. No wheeze means too little air is moving — escalate immediately.	
79	☐	Classic presentation of pulmonary embolism	Sudden breathlessness, pleuritic chest pain, tachycardia, sometimes haemoptysis. Think about it after immobility or surgery.	
80	☐	Chest drain — what does bubbling mean?	Continuous bubbling in the water seal suggests an air leak. Keep the unit upright and below chest level; never clamp for long.	
81	☐	What stays at a tracheostomy bedside?	The obturator, a spare tube of the same size, one a size smaller, and suction ready.	
82	☐	Positioning for breathlessness	Upright, high Fowler's or leaning forward on a table. Sitting up reduces the work of breathing before any drug does.	

Cardiac

5 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
83	☐	MONA — acute chest pain	Morphine · Oxygen · Nitrates · Aspirin. Relieve pain, then reassess — and get the ECG early.	
84	☐	Left versus right heart failure	Left backs up into the lungs: crackles, orthopnoea. Right backs up systemically: oedema, raised JVP, ascites.	
85	☐	Atrial fibrillation — the rhythm and the risk	Irregularly irregular, with no clear P waves. The real danger is clot formation and stroke, hence anticoagulation.	
86	☐	Adult chest compressions	Rate 100–120 per minute, depth 5–6 cm, 30:2 with ventilation, and full recoil between compressions.	
87	☐	Which chest pain is not cardiac?	Never assume. Pain reproducible on palpation or clearly positional is less likely cardiac, but an ECG and troponin settle it.	

Neuro

5 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
88	☐	Glasgow Coma Scale range	3 to 15. Eight or less means the airway is at risk — think intubation.	
89	☐	First action in a seizure	Protect the airway and the head, turn them to the side, and time it. Never force anything into the mouth.	
90	☐	Raised intracranial pressure — positioning	Head of bed elevated about 30°, head midline and neutral. Avoid neck flexion and hip flexion.	
91	☐	Stroke — why the time of onset matters	Thrombolysis is time-limited. The last time the patient was seen normal decides what treatment is possible.	
92	☐	The earliest sign of neurological deterioration	A change in level of consciousness. Pupil changes and vital sign changes come later.	

Maternal & Newborn

10 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
93	☐	When is APGAR scored, and out of what?	At 1 and 5 minutes. Five categories scored 0–2, so a maximum of 10.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
94	☐	How is preeclampsia defined?	New hypertension — BP $\geq$ 140/90 — after 20 weeks' gestation, with proteinuria or other organ involvement.	
95	☐	Preeclampsia with severe features	BP $\geq$ 160/110, severe headache, visual disturbance, epigastric or right upper quadrant pain. Escalate urgently.	
96	☐	First sign of magnesium sulfate toxicity	Loss of deep tendon reflexes, then respiratory depression. Stop the infusion; the antidote is calcium gluconate.	
97	☐	Late decelerations mean what?	Uteroplacental insufficiency. Reposition to the left side, give oxygen, stop oxytocin, increase IV fluid, and call for help.	
98	☐	Variable decelerations mean what?	Cord compression. Reposition first; consider amnioinfusion if they persist.	
99	☐	Early decelerations mean what?	Head compression during a contraction. They mirror the contraction and are benign.	
100	☐	Postpartum haemorrhage — the first action	Massage the fundus. Most early haemorrhage is uterine atony, and a boggy uterus is the clue.	
101	☐	Where should the fundus be after birth?	Firm, midline, at or just below the umbilicus. Boggy means atony; displaced to the side usually means a full bladder.	
102	☐	How much weight may a newborn lose?	Up to about 10% in the first week, regained by roughly two weeks. More than that needs review.	

Paediatrics

5 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
103	☐	Croup versus epiglottitis	Croup: barking cough, gradual. Epiglottitis: drooling, tripod posture, rapid onset — never inspect the throat, keep them calm.	
104	☐	Dehydration signs in an infant	Sunken fontanelle, no tears, dry mucosa, fewer wet nappies, lethargy. Weight is the most objective measure.	
105	☐	Never add medication to a full bottle of feed	If the infant does not finish it, the dose is unknown. Give it separately.	
106	☐	Why do children decompensate suddenly?	They compensate with tachycardia for a long time, then fall off quickly. Bradycardia in a sick child is pre-arrest.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
107	☐	Paediatric doses	Weight-based, and always double-checked. A decimal point error in paediatrics is a tenfold error.	

Mental Health

7 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
108	☐	A patient hints at suicide — what do you do?	Ask directly and specifically about intent, plan and means. Asking does not plant the idea; it establishes risk.	
109	☐	Neuroleptic malignant syndrome	Fever, lead-pipe rigidity, altered mental state, autonomic instability. Stop the antipsychotic — it is life-threatening.	
110	☐	Serotonin syndrome	Agitation, hyperreflexia, clonus, hyperthermia, often after combining serotonergic drugs. Stop the agents and escalate.	
111	☐	Lithium and hydration	Dehydration, salt restriction and NSAIDs all raise lithium levels. Steady fluid and salt intake keeps it safe.	
112	☐	Tardive dyskinesia	Involuntary repetitive movements, often of face and tongue, after long-term antipsychotics. It may not reverse — report early.	
113	☐	Therapeutic communication — what to avoid	Avoid 'why' questions, false reassurance, and giving advice. Use open questions, silence, and reflection.	
114	☐	Restraint principles	Last resort, least restrictive option, prescribed, time-limited, with documented monitoring and frequent review.	

Caribbean Focus

8 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
115	☐	Sickle cell crisis — the priorities	Hydration, oxygen, and adequate analgesia, usually opioid. Under-treating the pain is the commonest error.	
116	☐	What precipitates a sickle cell crisis?	Dehydration, hypoxia, cold, infection, acidosis and physical stress. Prevention teaching centres on fluids and avoiding cold.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
117	☐	Dengue warning signs	Severe abdominal pain, persistent vomiting, mucosal bleeding, lethargy, and a rising haematocrit with falling platelets.	
118	☐	Which analgesic in dengue?	Paracetamol. Avoid aspirin and NSAIDs — they worsen the bleeding risk.	
119	☐	Chikungunya — the distinguishing feature	Severe, often prolonged polyarthralgia alongside fever. Care is supportive; joint pain can persist for months.	
120	☐	Leptospirosis and flooding	Consider it after freshwater or floodwater exposure, especially with rodent contact. Fever with jaundice and renal impairment is the red flag.	
121	☐	Why is hypertension such a priority regionally?	It is highly prevalent in Caribbean populations and drives stroke, heart failure and chronic kidney disease. Adherence teaching is central.	
122	☐	Type 2 diabetes foot care teaching	Inspect daily including between the toes, never walk barefoot, well-fitting closed shoes, and never self-treat corns.	

Mnemonics

5 facts

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
123	☐	SBAR — handing over safely	Situation · Background · Assessment · Recommendation. It forces you to state what you actually want to happen.	
124	☐	SAMPLE — the rapid history	Signs · Allergies · Medications · Past history · Last intake · Events leading up.	
125	☐	PQRST — assessing pain	Provocation · Quality · Region and radiation · Severity · Timing.	
126	☐	RACE — fire response	Rescue anyone in danger · Alarm · Contain by closing doors · Extinguish or evacuate.	
127	☐	PASS — using an extinguisher	Pull the pin · Aim at the base · Squeeze · Sweep side to side.	

#	OK	PROMPT (FRONT OF CARD)	ANSWER (REVEAL)	CORRECTION / COMMENT
128	☐	Preventing falls — the highest-yield actions	Bed low and brakes on, call bell within reach, non-slip footwear, clear path to the toilet, and timed toileting rounds.	
129	☐	Before any procedure or transfusion	Two patient identifiers plus an independent check. Never use room number as an identifier.	
130	☐	A transfusion reaction begins — first action	Stop the transfusion immediately, keep the IV line with saline via fresh tubing, then assess and report.	
131	☐	When is a transfusion reaction most likely?	In the first 15 minutes. Stay with the patient and monitor closely through that window.	
132	☐	Documenting an error	Record what happened, the patient's condition, and who was notified — factually. Never document 'error' as an opinion or blame.	
133	☐	Informed consent — the nurse's role	The prescriber explains the procedure and risks. The nurse witnesses the signature and confirms the patient understands.	

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